

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

1. INTRODUCTION

1.1 Project Overview

This applied analytics project explores the intersection of dietary habits, academic performance, and lifestyle choices among college students. Using Tableau's advanced data visualization suite, raw survey profiles and transactional logs are transformed into actionable, interactive dashboards. The analysis uncovers critical patterns regarding skipping meals, academic stress, and eating behaviors.

1.2 Purpose

The project provides institutional stakeholders, nutritionists, and campus dining administrators with clear, evidence-backed insights to guide health interventions, structural wellness programs, and data-driven menu updates.

2. IDEATION PHASE

2.1 Problem Statement

College students frequently experience compromised physical well-being and reduced academic focus due to poor dietary habits, such as skipping breakfast, emotional stress-eating during exam periods, and high reliance on fast food. Simultaneously, campus cafeteria administrators lack the centralized data visibility required to align food options with actual student preferences and health needs.

2.2 Empathy Map Canvas

- **Says:** Students: "I skip breakfast and grab fast food because my engineering schedule is completely packed." "I find myself reaching for comfort food whenever exam anxiety kicks in." Cafeteria Manager: "We need clear metrics on student dietary trends to properly adjust our menu offerings."
- **Thinks:** "Does my diet have a measurable effect on my concentration or grades?" "Finding balanced, nutritious food on campus takes too much time during a busy day."
- **Does:** Consumes high-calorie junk food late at night, cuts out morning meals to save time, and experiences boredom-driven comfort food consumption.
- **Feels:** Sluggish, highly stressed, fatigued, and largely disconnected from true calorie awareness.

2.3 Brainstorming

- **Visual Indices:** Developing custom visual indices to connect meal consistency directly with concentration metrics.
- **Stress Tracking:** Isolating stress indicators and emotional triggers into dedicated charts to map comfort-food reliance.
- **Macro Segmentation:** Engineering macro-level canvas filters (such as Major and Residential Status) to find micro-trends among student demographics.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- **Discovery:** Students (e.g., Rahul or Priya) interact with the public Tableau dashboard during campus wellness workshops.
- **Insight Generation:** Users manipulate filters to see how skipping breakfast or experiencing high stress levels correlates with their focus drops or emotional eating dependencies.
- **Actionable Modification:** Students adjust their daily eating routines based on the data findings; concurrently, the cafeteria manager introduces salad bars and healthy combo options based on macro volume insights.

3.2 Solution Requirements

- **Functional Requirements:** System must clean raw survey data, handle missing stress/dietary records, perform transactional joins, build custom Focus/Stress indexes, and render dynamic dashboard filters.
- **Non-Functional Requirements:** Visual filtering actions must respond within minimal processing latency (<3 seconds) and support high canvas readability across distinct user groups.

3.3 Data Flow Diagram Summary

Raw Student Survey Data + Cafeteria Transaction Logs -> Tableau Prep (Join via Student ID) -> Missing Value Imputation -> Calculated Fields (Focus & Stress Indexes) -> Interactive Dashboards & Story Points -> End-User Decisions

3.4 Technology Stack

- **BI & Analytics Workspace:** Tableau Desktop / Tableau Cloud
- **Data Orchestration:** Tableau Prep Builder
- **Data Relational Layer:** Structured Flat Files (.xlsx / .csv) containing survey feedback and operational cafeteria transaction logs.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The dashboards address individual behaviors by making the consequences of diet visible—such as connecting skipped meals to drop-offs in focus. It simultaneously addresses administrative challenges by presenting macro-level consumption graphs that validate campus dining menu changes.

4.2 Proposed Solution

- **Student Insights Interface:** Visual workspaces mapping meal habits to concentration levels and tracing emotional stressors to comfort food reliance (Scenarios for Rahul and Priya).
- **Cafeteria Management Analytics Canvas:** A high-level view showing low fruit/vegetable intake against high fast-food volumes to guide food service updates.

4.3 Solution Architecture

- **Data Layer:** Raw student survey tables merged with historical cafeteria transaction data.
- **Semantic Analytics Layer:** Custom Tableau logical statements computing the Focus Index and Stress Index.
- **Presentation Layer:** Dynamic dashboards equipped with interactive action filters and cohesive case-study narrative layouts.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The development lifecycle followed an Agile Scrum framework, executing a total scope of 22 Story Points across a scheduled 12-day timeline split into 3 distinct sprints. Average Velocity (AV) per day is calculated as follows:

$$AV \text{ per Day} = \text{Total Story Points} / \text{Total Project Duration} = 22 \text{ Points} / 12 \text{ Days} = 1.83 \text{ Story Points / Day}$$

$$\text{Sprint Velocity} = 22 \text{ Story Points} / 3 \text{ Sprints} = 7.33 \text{ Story Points per Sprint}$$

Sprint & Backlog Details Table

Sprint	Epic / Requirement	USN	User Story / Task	Points	Priority
Sprint-1	Data Pipeline Setup & Clean	USN-1	Clean raw student survey dataset to fix missing values.	3	High
Sprint-1	Data Pipeline Setup & Clean	USN-2	Join survey responses with cafeteria transaction logs.	2	High

Sprint-1	Metric Frameworks	USN-3	Build Tableau calculated fields for focus and stress indexes.	2	Medium
Sprint-2	Student Insights View	USN-4	Interactive dashboard linking skipped breakfast to focus drop.	5	High
Sprint-2	Student Insights View	USN-5	Trend lines mapping emotional stressors to comfort food.	3	High
Sprint-3	Cafeteria Admin View	USN-6	Canvas showing low produce vs high fast food trends.	5	High
Sprint-3	Cafeteria Admin View	USN-7	Filter student configuration by major/residential status.	2	Medium

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing Metrics

Model and dashboard testing information logged to ensure seamless UI responsiveness:

- **Data Rendered:** Successfully joined primary student surveys with transactional cafeteria logging tables via student primary keys.
- **Preprocessing Verification:** Standardized data formatting and imputed empty attributes within critical dietary parameters.
- **Utilization of Filters:** Verified global interactive filters (Major, Residential Status) and cross-highlighting actions with latency < 3 seconds.
- **Dashboard & Story Design:** 5 functional visualizations and 3 interlinked interactive case-study story points completely validated.

7. RESULTS

7.1 Output Framework and Scenarios

- **Scenario 1 (Rahul's Case):** The dashboard links skipped breakfast routines directly to concentration drop-offs, driving better health and nutrition choices.
- **Scenario 2 (Priya's Case):** The interface maps exam stress and boredom indicators directly to comfort food dependencies to assist personal stress management.

- **Scenario 3 (Cafeteria Admin View):** Provides high-level visual data of low produce intake vs high fast food volume to configure better regional menus.

8. ADVANTAGES & DISADVANTAGES

- **Advantages:** Establishes data-driven evidence connecting nutrition to academic focus; enables quick cafeteria menu optimization; highly interactive and user-friendly.
- **Disadvantages:** Relies on self-reported survey parameters which can introduce subjective bias; insights depend entirely on regular transactional log updates to remain relevant.

9. CONCLUSION

The Tableau workbook successfully bridges the gap between student lifestyle tracking and institutional dining administration. By contextualizing behavioral data into analytical narratives, the case study provides clear evidence that structured eating habits improve student performance and guides universities toward healthier campus environment configurations.

10. FUTURE SCOPE

- **Predictive Modeling:** Integrating predictive modeling to flag students at high risk for extreme stress-eating based on historical exam calendars.
- **Live Data Automation:** Automating live data connections from cafeteria point-of-sale systems to dynamically update dashboard inventory insights.
- **Mobile Companion:** Developing a mobile-responsive companion view for students to log personal calorie choices and see real-time wellness impacts.

11. APPENDIX

- **Dataset Link:** <https://university-portal.edu/datasets/student-dietary-habits-2026.csv>
- **GitHub & Project Demo Link:** <https://github.com/yashwitha-nara/tableau-student-dietary-analysis>